End-to-End Backend Service for Wine Country Prediction

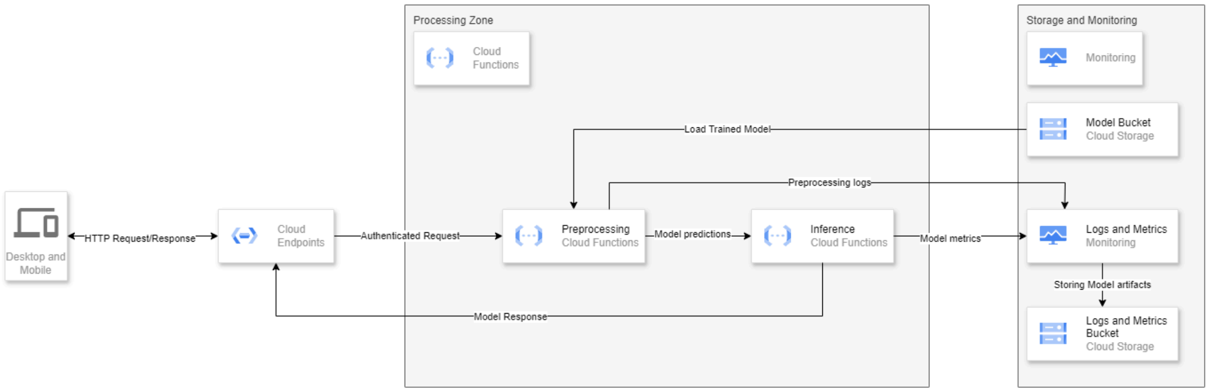
1. Architecture Overview

In this report, we will discuss 2 proposed architectures for ML model that predicts the country of origin of a wine. The approaches are categorised as follows:

- **Cost-Efficient Workflow**: A serverless, pay-as-you-go model designed for small to medium workloads.
- **Standard Production-Grade Workflow**: A scalable and high-performance architecture for complex and large-scale use cases.

Both workflows have cloud-based preprocessing, model predictions, monitoring, and data storage components. The key differences lie in the preprocessing and model deployment.

2. Cost-Efficient Workflow



Architecture Flow:

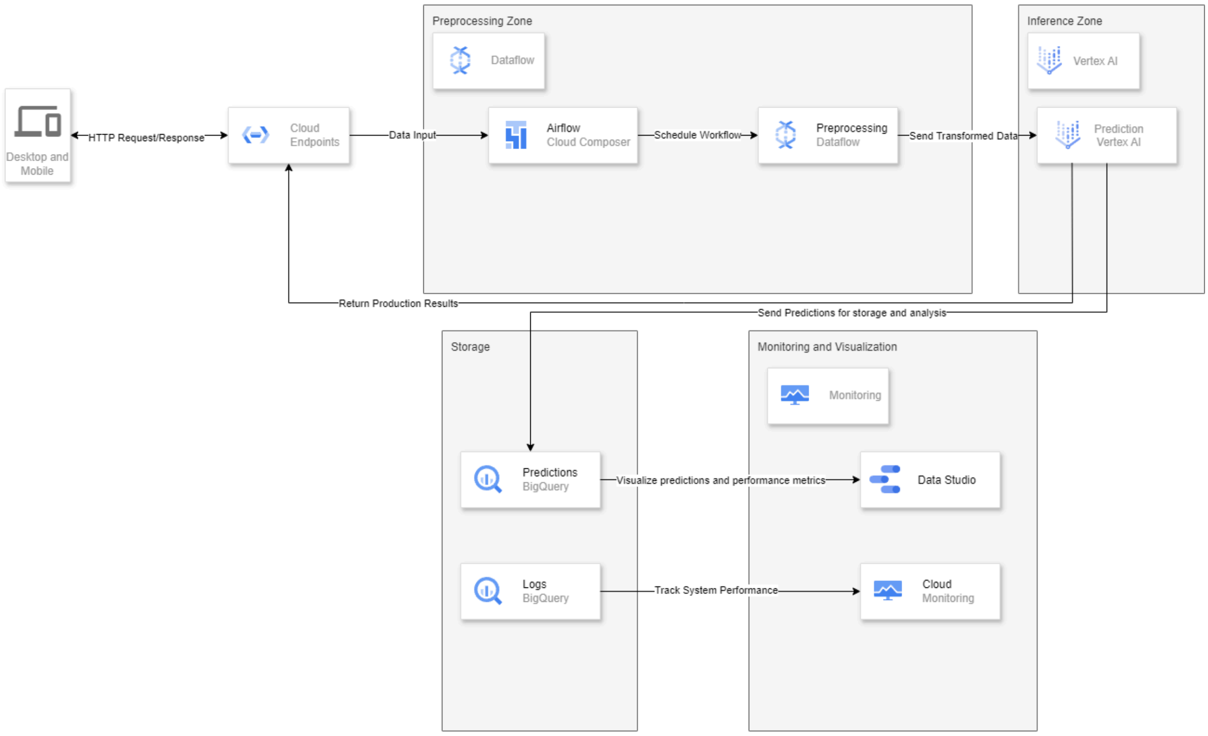
- **Client Interaction:**
  - A client submits an HTTP request containing wine review data (description, price, points, and variety).
- **API Gateway - Cloud Endpoints:**
  - Routes and authenticates the request before processing.
- **Preprocessing with Cloud Functions:**
  - Performs lightweight data transformations (e.g., tokenizing descriptions, normalizing numerical features).
- **Model Loading & Inference:**
  - Cloud Functions loads the trained ML model from Google Cloud Storage (GCS).
  - The function runs the model and predicts the country of origin.
- **Returning the Response:**
  - The processed prediction result is routed back through Cloud Endpoints to the client.

- **Monitoring & Logging:**
    - Logs and system performance metrics are captured using Cloud Monitoring for tracking API calls and function execution times.

Justifications

- **Serverless**: Reduces operational overhead and cost by using Cloud Functions.
- **Simple and Efficient**: Suitable for applications when there are not a lot of requests.
- **Scalability Constraints**: Since Cloud Functions have execution time limits, this approach is not ideal for large-scale workloads.

3. Standard (Production-Grade) Workflow



Architecture Flow:

- **Client Interaction:**
    - A client sends an HTTP request with wine review data.
- **API Gateway - Cloud Endpoints:**
    - Routes and authenticates the request before processing.
- **Preprocessing with Dataflow (Optional):**
    - Cloud Composer with Airflow DAGs is used to orchestrate preprocessing workflows.
    - Cloud Dataflow performs scalable data transformations, such as text vectorization and normalization.
- **Model Inference with Vertex AI Prediction:**
    - The preprocessed data is sent to Vertex AI Prediction, which hosts the trained ML model.
    - The model processes the input and returns a prediction.
- **Data Storage in BigQuery:**

- Predictions and logs are stored in BigQuery for analysis and reporting.
- **Returning the Response:**
  - The final prediction result is sent back through Cloud Endpoints to the client.
- **Monitoring, Logging & Tracing:**
  - Cloud Monitoring captures system performance metrics.
  - Data Studio visualizes predictions and performance trends.

Justifications

- **Scalability**: Ideal for high-traffic applications requiring continuous predictions.
- **Data Processing Power**: Cloud Dataflow efficiently preprocesses large datasets.
- **Optimized Model Hosting**: Vertex AI provides model versioning, auto-scaling, and optimized GPU serving.
- **Advanced Monitoring**: Integration with BigQuery and Data Studio allows for deep analytics and model performance tracking.

4. Deployment Strategy

Deployment Steps

- **Infrastructure as Code (IaC):**
  - Use Terraform to automate cloud resource provisioning.
- **CI/CD Pipeline:**
  - Set up a continuous integration and deployment with GitHub Actions.
- **Model Deployment:**
  - Train models in Vertex AI Training and deploy them to Vertex AI Prediction.
  - Store trained models in Google Cloud Storage (GCS).
- **Workflow Automation:**
  - Use Cloud Composer (Airflow DAGs) to automate data preprocessing and training pipelines.
- **Monitoring and Optimization:**
  - Implement Cloud Monitoring and Logging to track system performance.
  - Conduct periodic model retraining and hyperparameter tuning to maintain accuracy. (Can be done using TFX libraries to schedule retraining based on time/data or other conditions)

Reasoning

- **Automated & Repeatable**: Infrastructure as Code (IaC) ensures consistency and reliability.
- **Efficient Model Updates**: CI/CD pipelines allow continuous iteration and deployment.
- **Cost vs. Performance Trade-off**: The two workflows provide flexibility based on demands.
- **Scalable Architecture**: The standard workflow can handle increasing demand without downgrading performance.

5. AWS Alternatives

GCP Component	AWS Alternative
Cloud Endpoints	Amazon API Gateway
Cloud Functions	AWS Lambda
Google Cloud Storage (GCS)	Amazon S3
Cloud Monitoring	Amazon CloudWatch
Vertex AI Prediction	Amazon SageMaker Endpoint
Cloud Run	AWS Fargate / ECS
Vertex AI Training	Amazon SageMaker Training
Dataflow	AWS Glue / AWS Data Pipeline
Cloud Composer (Airflow)	Amazon Managed Workflows (MWAA)
BigQuery	Amazon Redshift / Athena
Vertex AI Model Registry	Amazon SageMaker Model Registry
Vertex AI Pipelines	Amazon SageMaker Pipelines
Pub/Sub	Amazon SNS / Amazon Kinesis
Deployment Manager	AWS CloudFormation
Cloud Build	AWS CodePipeline

Notes:

- **Amazon SageMaker** is the closest equivalent to **Vertex AI**, providing model training, deployment, and inference.
- **AWS Glue** serves as a parallel for **Dataflow** in handling large-scale ETL and preprocessing.
- **AWS Lambda** is used in place of **Cloud Functions** for serverless processing.

6. Conclusion

The choice between the Cost-Efficient Workflow and the Standard Workflow depends on the specific requirements:

- For smaller workloads, the cost-efficient approach is preferable due to its low maintenance and pay-as-you-go pricing.
- For enterprise-grade solutions, the standard workflow provides better scalability, performance, and observability.

Both solutions leverage Google Cloud’s serverless architecture, managed ML services, and automated pipelines to deliver machine learning backend. The system ensures high availability, low latency, and efficient inference serving at scale.